Lecture 1. Introduction to International Trade

Lectures on International Trade

Instructor: Xiaoping Chen

Harbin Institute of Technology, Shenzhen

Preview

- What is international trade about?
  - Goods, services, labor and investment
  - Role of trade in the economy
  - Trade flows: a gravity equation
- Key topics in international trade
  - Gains from trade
  - Specialization patterns
  - Trade policies
  - Other advanced topics

What is international trade about?

- Trade of goods
  - Final goods
  - Intermediate inputs
- Trade of services
  - Used to be non-tradable
  - More and more important

What is international trade about?

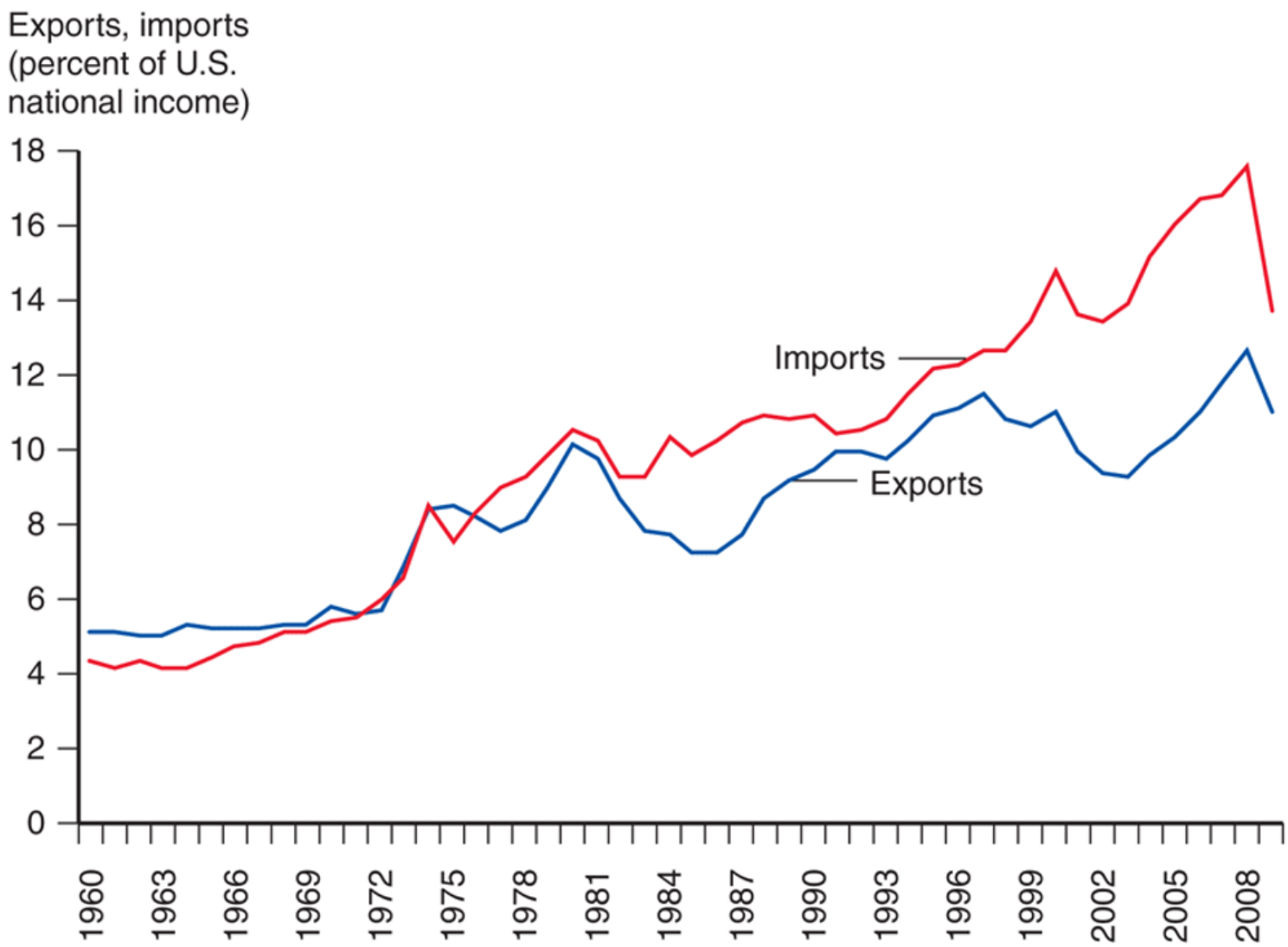
- Trade of goods
  - Final goods
  - Intermediate inputs
- Trade of services
  - Used to be non-tradable
  - More and more important
- Immigration
- FDI

What is international trade about? (cont.)

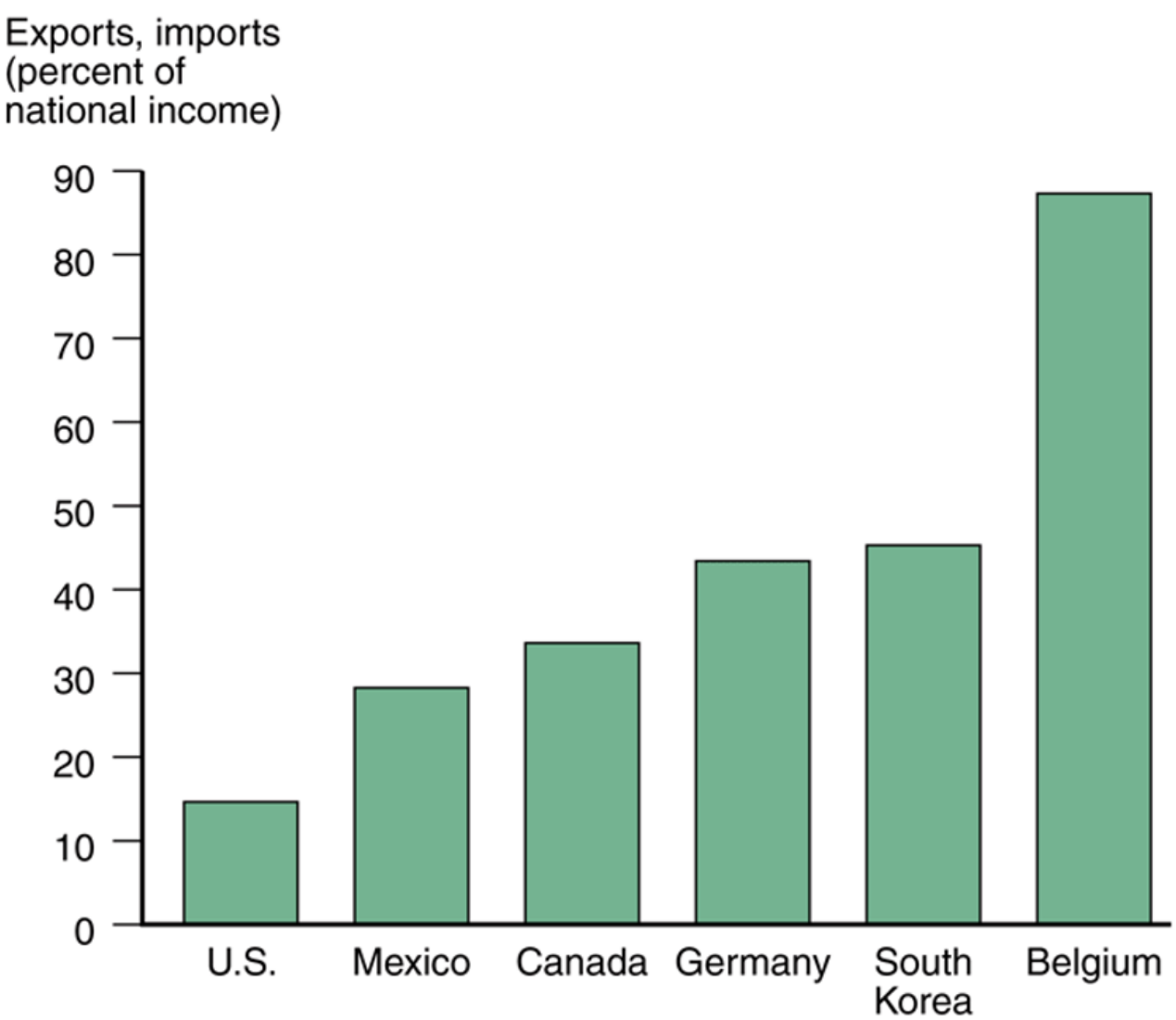
Trade as a fraction of GDP

- International trade as a fraction of the national economy has tripled for the U.S. in the past 40 years.
- Compared to the U.S., other countries are even more tied to international trade.

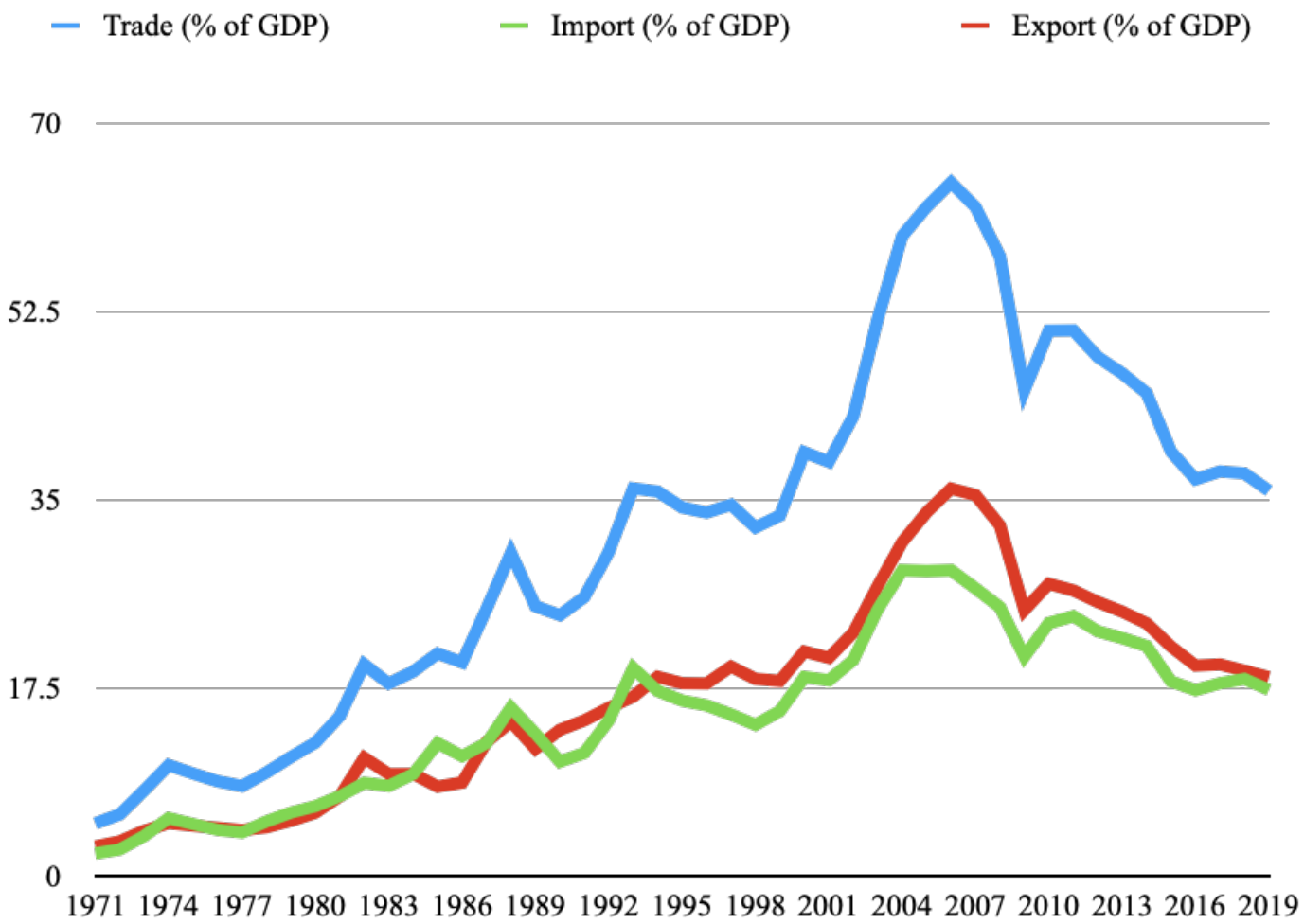
What is international trade about? (cont.)



What is international trade about? (cont.)



What is international trade about? (cont.)



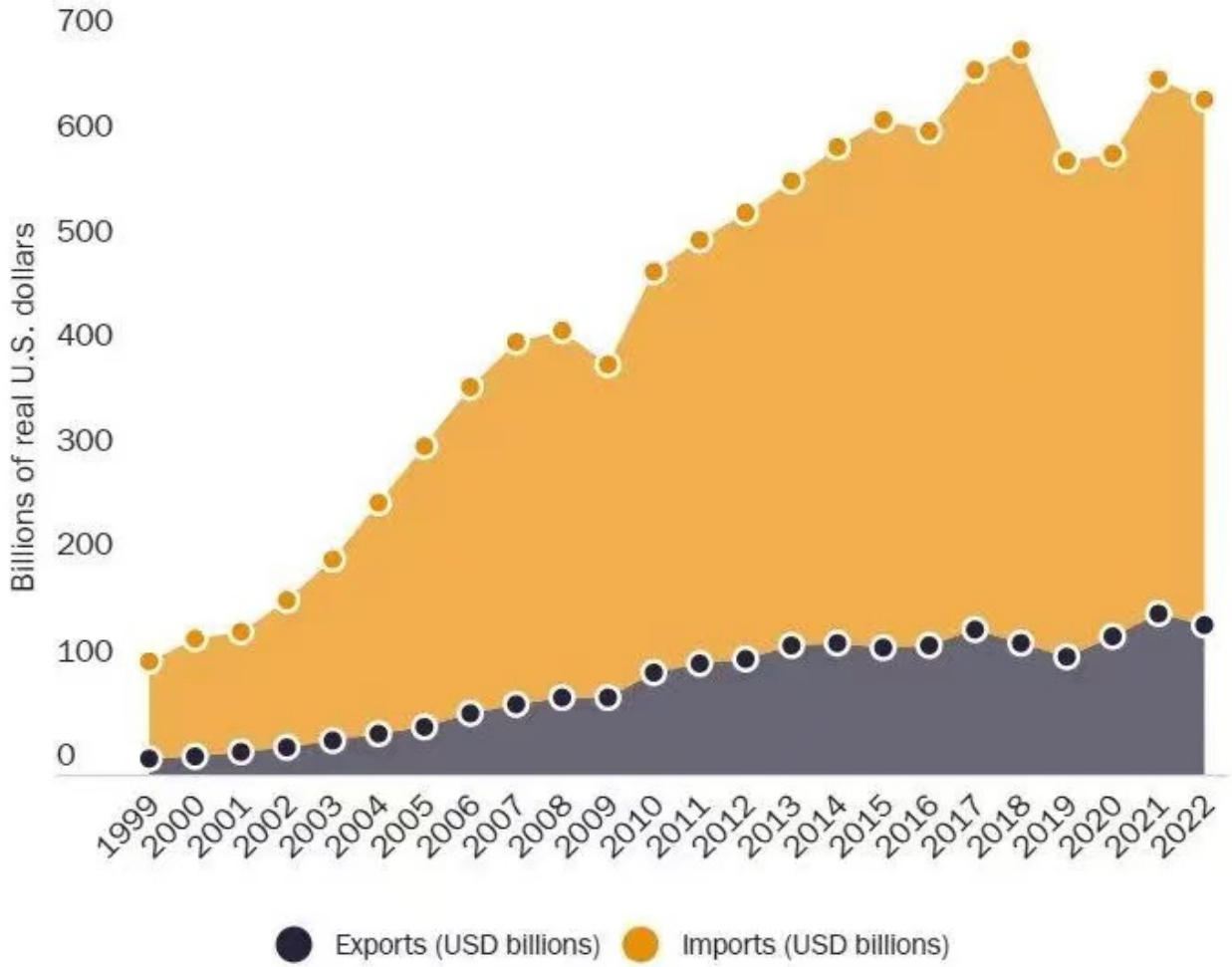
Trade (Import and Export) in China (% of GDP)

Source: World Development Indicators from World Bank Database.

What is international trade about? (cont.)

Figure 1

Inflation-adjusted U.S. trade in goods with China remained high in 2022



Sources: Nominal data from U.S. Census Bureau, adjusted for inflation using the Bureau of Economic Analysis' Personal Consumption Expenditures index for goods.

What is international trade about? (cont.)

A Gravity Equation of Trade

- Gravity equation in physics

$$F = G\frac{m_1 \cdot m_2}{r^2}.$$

What is international trade about? (cont.)

A Gravity Equation of Trade

- Gravity equation in physics

$$F = G\frac{m_1 \cdot m_2}{r^2}.$$

- Gravity equation in trade

$$X_{ij} = C\frac{Y_i Y_j}{\tau_{ij}^{\theta}}.$$

What is international trade about? (cont.)

A Gravity Equation of Trade

- Gravity equation in physics

$$F = G\frac{m_1 \cdot m_2}{r^2}.$$

- Gravity equation in trade

$$X_{ij} = C\frac{Y_i Y_j}{\tau_{ij}^{\theta}}.$$

- Trade costs
  - Geography; Transportation technology
  - Trade policies
  - Language; Culture; Natural disaster

Gains from Trade

Several ideas underlie the gains from trade

1. When a buyer and a seller engage in a voluntary transaction, both can be made better off.
   - Norwegian consumers import oranges that they would have a hard time producing.
   - The producer of the oranges receives income that it can use to buy other things that it desires.
2. How could a country that is the most (least) efficient producer of everything gain from trade?
   - Countries use finite resources to produce what they are most productive at (compared to their other production choices), then trade those products for goods and services that they want to consume.
   - Countries can specialize in production, while consuming many goods and services through trade.

Gains from Trade (cont.)

3. Trade benefits countries by allowing them to export goods made with relatively abundant resources and import goods made with relatively scarce resources.
4. When countries specialize, they may be more efficient due to larger-scale production.
5. Countries may also gain by trading current resources for future resources (international borrowing and lending) and due to international migration.
6. Trade usually increases market competition and decreases distortion.

Gains from Trade (cont.)

- Trade is predicted to benefit countries as a whole in several ways, but trade may harm particular groups within a country.
  - International trade can harm the owners of resources that are used relatively intensively in industries that compete with imports.
  - Trade may therefore affect the distribution of income within a country.
- Gains are compared to itself without trade, not compared to others.

Specialization Patterns of Trade

- Differences in climate and resources can explain why Brazil exports coffee and Australia exports iron ore.
- But why does Japan export automobiles, while the U.S. exports aircraft?
- Comparative advantage
- Why some countries export certain products can stem from differences in:
  - Labor productivity
  - Relative supplies of capital, labor and land and their use in the production of different goods and services
  - Institutions
  - Other sources of comparative advantage

Effects of Government Policies on Trade

- Policy makers affect the amount of trade through
  - tariffs: a tax on imports or exports,
  - quotas: a quantity restriction on imports or exports,
  - export subsidies: a payment to producers that export,
  - or through other regulations (ex., product specifications) that exclude foreign products from the market, but still allow domestic products.
- What are the costs and benefits of these policies?

Effects of Government Policies on Trade (cont.)

- If a government must restrict trade, which policy should it use and how much should it restrict trade?
- If a government restricts trade, what are the costs if foreign governments respond likewise?
- Trade policies are often chosen to cater to special interest groups, rather than to maximize national welfare.
- Governments tend to adopt tariffs, then negotiate them down in exchange for reduction in trade barriers of other countries.
- Other strategic trade policies and non-tariff trade barriers.